Haoyi Zhu

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EDUCATION

University of Science and Technology of China (USTC), China *Sep. 2023 – Present*
Ph.D. in Computer Science (Advisor: Prof. Tong He & Prof. Chunhua Shen)
Shanghai Jiao Tong University (SJTU), China *Sep. 2019 – Jun. 2023*
B.E. in Artificial Intelligence Honor Class (Advisor: Prof. Cewu Lu)

SELECTED AWARDS

China3DV Top 1 Best Paper Award *Apr. 2026*
China3DV Academic Rising Star Award Candidate *Apr. 2026*
Outstanding Paper Award, ICCV 2025 RIWM Workshop *Oct. 2025*
National Scholarship *Oct. 2025*
First Price Academic Scholarship, USTC *Sep. 2025*
Outstanding Paper Award, NeurIPS 2022 *Dec. 2022*

INTERNSHIPS

ATH, Alibaba *Jul. 2026 - Present*
Research Intern (Advisor: Dr. Xinlong Wang)
- Video Generation
- Sparse Linear Attention

Efficient AI Lab, NVIDIA *Dec. 2025 - Jul. 2026*
Research Intern (Advisor: Dr. Enze Xie & Prof. Song Han)
- World Model

MVIG, Shanghai Jiao Tong University (SJTU) *Sep. 2019 - Jun. 2023*
Research Intern (Advisor: Dr. Hao-Shu Fang & Prof. Cewu Lu)
- Human Pose Estimation
- General Robot Manipulation

GEAR Lab, NVIDIA *Feb. 2022 - Feb. 2023*
Research Intern (Advisor: Dr. Guanzhi Wang & Dr. Jim Fan)
- General Open-Ended Embodied Agent

Xu Lab, Carnegie Mellon University (CMU) *Apr. 2021 – Feb. 2022*
Research Intern (Advisor: Prof. Min Xu)
- AI for Biology

PUBLICATIONS

Total Citations: 3949

Journal Papers

PonderV2: Pave the Way for 3D Foundation Model with A Universal Pre-training Paradigm [paper] [code]
Haoyi Zhu*, Honghui Yang*, Xiaoyang Wu*, Di Huang*, Sha Zhang, Xianglong He, Tong He, Hengshuang Zhao, Chunhua Shen, Yu Qiao, Wanli Ouyang
*IEEE Transactions on Pattern Analysis and Machine Intelligence (**TPAMI**), 2025*

AlphaTracker: A Multi-Animal Tracking and Behavioral Analysis Tool [paper] [code]
Zexin Chen, Ruihan Zhang, Hao-Shu Fang, Yu E Zhang, Aneesh Bal, Haowen Zhou, Rachel R Rock, Nancy Padilla-Coreano, Laurel R Keyes, **Haoyi Zhu**, Yong-Lu Li, Takaki Komiyama, Kay M Tye, Cewu Lu
Frontiers in Behavioral Neuroscience, 2023

AlphaPose: Whole-Body Regional Multi-Person Pose Estimation and Tracking in Real-Time [paper] [code]
Hao-Shu Fang, Jiefeng Li, Hongyang Tang, Chao Xu*, **Haoyi Zhu***, Yuliang Xiu, Yong-Lu Li, Cewu Lu
[>8.6K GitHub Stars] *IEEE Transactions on Pattern Analysis and Machine Intelligence (**TPAMI**), 2022*

Conference Papers (Selected)

SANA-WM: Efficient Minute-Scale World Modeling with Hybrid Linear Diffusion Transformer [paper] [website] [code]
Haoyi Zhu, Haozhe Liu, Yuyang Zhao, Tian Ye, Junsong Chen, Jincheng Yu, Tong He, Song Han, Enze Xie
[>9.3K GitHub Stars] *arXiv preprint, 2026*

OmniWorld: A Multi-Domain and Multi-Modal Dataset for 4D World Modeling [paper] [website] [code]
..., **Haoyi Zhu**, ... (19 authors)
*International Conference on Learning Representations (**ICLR**), 2026*

WinT3R: Window-Based Streaming Reconstruction with Camera Token Pool [paper] [website] [code]
Yifan Wang, Jianjun Zhou, **Haoyi Zhu**, Wenzheng Chang, Yang Zhou, Zizun Li, Junyi Chen, Jiangmiao Pang, Chunhua Shen, Tong He
*International Conference on Learning Representations (**ICLR**), 2026*

π^3: Scalable Permutation-Equivariant Visual Geometry Learning [paper] [website] [code]
Yifan Wang, Jianjun Zhou, **Haoyi Zhu**, Wenzheng Chang, Yang Zhou, Zizun Li, Junyi Chen, Jiangmiao Pang, Chunhua Shen, Tong He
[>2.2K GitHub Stars & Top-1 Best Paper at China3DV] *International Conference on Learning Representations (**ICLR**), 2026*

DeepVerse: 4D Autoregressive Video Generation as a World Model [paper] [website] [code]
Junyi Chen, **Haoyi Zhu**, Xianglong He, Yifan Wang, Jianjun Zhou, Wenzheng Chang, Yang Zhou, Zizun Li, Zhoujie Fu, Jiangmiao Pang, Tong He
arXiv preprint, 2025

VQ-VLA: Improving Vision-Language-Action Models via Scaling Vector-Quantized Action Tokenizers [paper] [website][code]
Yating Wang, **Haoyi Zhu**, Mingyu Liu, Jiange Yang, Hao-Shu Fang, Tong He
*International Conference on Computer Vision (**ICCV**), 2025*

Aether: Geometric-Aware Unified World Modeling [paper] [website] [code]
Haoyi Zhu, Yifan Wang, Jianjun Zhou, Wenzheng Chang, Yang Zhou, Zizun Li, Junyi Chen, Chunhua Shen, Jiangmiao Pang, Tong He
*International Conference on Computer Vision (**ICCV**), 2025*
[Outstanding Paper Award & Oral Presentation] *ICCV 2025 Workshop on Reliable and Interactive World Models (**RIWM**)*

CoMo: Learning Continuous Latent Motion from Internet Videos for Scalable Robot Learning [paper]
Jiange Yang, Yansong Shi, **Haoyi Zhu**, Mingyu Liu, Kaijing Ma, Yating Wang, Gangshan Wu, Tong He, Liming Wang
*IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2026*

SPA: 3D Spatial-Awareness Enables Effective Embodied Representation [paper] [website] [code]
Haoyi Zhu, Honghui Yang, Yating Wang, Jiange Yang, Limin Wang, Tong He
*International Conference on Learning Representations (**ICLR**), 2025*

Tra-MoE: Learning Trajectory Prediction Model from Multiple Domains for Adaptive Policy Conditioning [paper] [code]
Jiange Yang, **Haoyi Zhu**, Yating Wang, Gangshan Wu, Tong He, Limin Wang
*IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2025*

Point Cloud Matters: Rethinking the Impact of Different Observation Spaces on Robot Learning [paper] [website] [code]
Haoyi Zhu, Yating Wang, Di Huang, Weicai Ye, Wanli Ouyang, Tong He
*Advances in Neural Information Processing Systems (**NeurIPS**), 2024*

RH20T: A Comprehensive Robotic Dataset for Learning Diverse Skills in One-Shot [paper] [website] [code]
Hao-Shu Fang, Hongjie Fang, Zhenyu Tang, Jirong Liu, Chenxi Wang, Junbo Wang, **Haoyi Zhu**, Cewu Lu
*IEEE International Conference on Robotics and Automation (**ICRA**), 2024*

UniPAD: A Universal Pre-Training Paradigm for Autonomous Driving [paper] [code]
Honghui Yang, Sha Zhang, Di Huang, Xiaoyang Wu, **Haoyi Zhu**, Tong He, Shixiang Tang, Hengshuang Zhao, Qibo Qiu, Binbin Lin, Xiaofei He, Wanli Ouyang
*IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2024*

X-NeRF: Explicit Neural Radiance Field for Multi-Scene 360° Insufficient RGB-D Views [paper] [code]
Haoyi Zhu, Hao-Shu Fang, Cewu Lu
*IEEE/CVF Winter Conference on Applications of Computer Vision (**WACV**), 2023*

MineDojo: Building Open-Ended Embodied Agents with Internet-Scale Knowledge [paper] [website] [code]
Linxi Fan, Guanzhi Wang, Yunfan Jiang, Ajay Mandlekar, Yuncong Yang, **Haoyi Zhu**, Andrew Tang, De-An Huang, Yuke Zhu, Anima Anandkumar
[Outstanding Paper Award] *Advances in Neural Information Processing Systems (**NeurIPS**), 2022*

For the full publication list, please visit my Google Scholar page.

BLOG POSTS

Diffusion Notes: When Are ODE and SDE Sampling Equivalent?	*Sep. 2026*
Sparse Linear Attention: When Sparsity Meets Linear Attention (2)	*Aug. 2026*
Reflections on Video DeltaNet	*Aug. 2026*
Sparse Linear Attention: When Sparsity Meets Linear Attention	*Jul. 2026*

SELECTED PROJECTS

SANA-WM

Efficient Minute-Scale World Modeling with Hybrid Linear Diffusion Transformer — *Dec. 2025 – Present*

- Developed **SANA-WM**, a **2.6B-parameter open-source** world model that turns one image, text, and a camera trajectory into **60-second 720p** controllable videos with precise 6-DoF camera control on a single GPU; the Sana codebase has **>9.3K GitHub stars**.
- Built the efficient one-minute backbone by pairing an LTX2 high-compression tokenizer with **hybrid linear attention**: 15 frame-wise Gated DeltaNet blocks for recurrent long-context modeling and 5 softmax-attention blocks for exact long-range recall.
- Designed **dual-branch camera control**, combining a latent-rate UCPE branch for global trajectory structure with raw-frame Plücker ray mixing for fine camera motion within each temporal VAE stride.
- Built the data and deployment pipeline: recovered metric-scale 6-DoF poses from public videos to train on $\sim$**200K** annotated clips, completed training in 15 days on 64 H100s, and used a long-video refiner/distilled variant to reach competitive visual quality, stronger action following, and up to **36**$\times$ **higher throughput**.
- Released an **NVFP4-distilled streaming** final version for interactive minute-duration generation, running at $\sim$**23 FPS** on a single RTX 5090 GPU.

Aether

Scalable and Generalizable Foundation World Model — *Nov. 2024 – Nov. 2025*

- Proposed **Aether**, a geometry-aware unified world model. **Aether** features 3 core capabilities: (1) 4D dynamic reconstruction, (2) action-conditioned video prediction, and (3) goal-conditioned visual planning. Building upon video generation models, our framework demonstrates unprecedented synthetic-to-real generalization despite *never* observing real-world data during training. Won the **Outstanding Paper Award** at ICCV 2025 RIWM workshop.
- Developed **DeepVerse**, a 4D auto-regressive world model. **DeepVerse** can fantasize the entire world behind images and enable free exploration through interaction. It demonstrates exceptional capabilities in modeling dynamics, geometry, and physical laws.
- Introduced π^3, a novel feed-forward neural network that revolutionizes visual geometry reconstruction by eliminating the need for a fixed reference view. With a fully permutation-equivariant architecture, π^3 achieves SOTA performance on a wide range of tasks, including camera pose estimation, monocular/video depth estimation, and dense point map estimation.
- Mentored several undergraduate and junior Ph.D. students on robot learning topics, including action tokenizers, cross-embodiment policies, point cloud policies, *etc.*

MineDojo

Building Open-Ended Embodied Agents with Internet-Scale Knowledge — *Feb. 2022 – Jul. 2022*

- Contributed to the ***MineDojo*** project, a new framework designed to build generally capable agents with internet-scale knowledge in Minecraft. Won the **Outstanding Paper Award** at NeurIPS 2022.
- Co-developed the YouTube database, which comprises over 730,000 narrated Minecraft videos, totaling approximately 300,000 hours of content and 2.2 billion English transcripts.
- Constructed MineCLIP, a video CLIP foundation model that acts as a learned reward function, enabling agents to solve diverse open-ended tasks specified in natural language.
- Authored APIs for three databases (YouTube, Minecraft Wiki, and Reddit), along with the corresponding documentation and part of the official project website.

ACADEMIC SERVICES

Reviewer for journals, including IJCV, TMLR, *etc*.
Reviewer for conferences, including ICCV, CVPR, NeurIPS, ICLR, WACV, ICIP, *etc*.

INVITED TALKS

- 2025/08/26, Prof. Lin Shao's Lab, NUS, "World Model and Spatial Intelligence"
- 2025/06/12, 3D CV Workshop (3D视觉工坊), "Towards Generalizable and Scalable Spatial Embodied Intelligence" [link]
- 2025/05/22, Embodied Intelligent Mind (具身智能之心), "Towards Generalizable and Scalable Spatial Embodied Intelligence"
- 2025/04/22, Prof. Yanyong Zhang's Lab, USTC, "Towards Scalable World Model for General Embodied AI"
- 2025/03/27, Graduate Academic Forum, USTC, "Towards Spatial Intelligence: From 3D Vision to Embodied AI"
- 2024/12/27, ZhiXingXing Embodied AI Frontier Lecture (智猩猩具身智能前沿讲座), "Towards Spatial Intelligence: From 3D Vision to Embodied AI" [link]